

Roger Chan

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EDUCATION

University of Waterloo, Faculty of Engineering <i>Master of Management Engineering (MMSc), Specialization in Data Science</i>	Waterloo, ON <i>Expected Sept 2027</i>
University of Waterloo, Faculty of Health <i>Bachelor of Science, Minor in Computing</i>	Waterloo, ON <i>Graduated 2024</i>

EXPERIENCE

Data Science Intern <i>Wealthsimple</i>	Jan 2026 – Present <i>Toronto, ON</i>
- Tuned class-weight dampening and regularization on a PyTorch foundation model (46M params, 6-layer transformer) trained on 8.1M financial event sequences, improving next-event prediction accuracy from 70.3% to 75.0% and macro F1 by 4.7% across 2.55M user embeddings used for fraud detection. - Built an LLM-powered AML alert triage system using Claude (AWS Bedrock) that achieved an 87% match rate against analyst dispositions across 1,805 cases, reducing dangerous false dismissals to 0.27%. - Built a dbt pipeline monitoring bot (custom Airflow operator + Slack) that streams live progress for multi-hour data-mart builds; engineered its cross-pod logging on Kubernetes via shared EFS storage to carry it through the platform's Nomad-to-K8s migration, and shipped it as the first cross-team adopter. - Audited 240+ Preset datasets across the market-risk & finance schema and consolidated a near-duplicate cluster, 1,800+ lines of copy-pasted SQL) into a single governed dbt model as the team's source of truth for margin-stress risk reporting; validated parity within 0.07% and caught a divergent legacy pipeline before archiving.	
Data Science Intern <i>Vale Base Metals</i>	May 2024 – Dec 2025 <i>Toronto, ON</i>
- Developed and maintained scalable data pipelines in Python and Power Automate to centralize Continuous Improvement data from four global regions, ensuring data quality and governance. - Integrated classification ML models into PowerApps to recommend optimal CI methodologies, automating project routing and boosting submission throughput by 25%.	
Data Science Intern <i>Ontario Securities Commission (OSC)</i>	May 2023 – Dec 2023 <i>Toronto, ON</i>
- Built ETL pipelines in Python to extract and normalize financial disclosure data from thousands of unstructured PDFs, reducing manual extraction time by 20%. - Applied NLP models (spaCy, Transformers) to identify and classify fund-related entities, improving text extraction accuracy across millions of regulatory records. - Collaborated with data engineers and auditors to automate validation processes using SQL-based transformations, cutting audit processing time by 60%.	

PROJECTS

City of Iqaluit Smart Water Delivery App (Capstone)	Jan 2024 – Apr 2024
- Co-developed a mobile application with the City of Iqaluit to modernize trucked water delivery for 470+ households, reducing operational costs by \$250K/year. - Integrated IoT sensors with Firebase to enable real-time water tank monitoring and automated delivery scheduling. - Implemented an ARIMA-based ML forecasting model to predict household water demand and minimize off-schedule deliveries.	

TECHNICAL SKILLS

Languages: Python, SQL (Redshift, PostgreSQL), R, Bash/Shell, JavaScript, HTML, CSS
AI / ML: PyTorch, Transformers, Claude API (AWS Bedrock), Claude Code (Agentic Skills), Prompt Engineering, Scikit-learn, spaCy, LangChain
Data & Visualization: dbt, Preset (Apache Superset), Power BI, Tableau, Pandas, NumPy, Polars, Matplotlib, ggplot2
Tools & Platforms: AWS (Bedrock, EFS), Azure, Apache Airflow, Kubernetes, Nomad, DVC, MLflow, Git, Jupyter
Core Competencies: AML/KYC Compliance, Data Modeling, ETL & Pipeline Orchestration, Pipeline Monitoring, LLM Integration